Agrilink

Empowering Farmers

Through smart technology

A Capstone Project

Presented to the Faculty of the

College of Information Technology Education

PHINMA University of Iloilo

In Partial Fulfillment

of the Requirements for the degree

Bachelor of Science in Information Technology

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COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

Approval Sheet

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Abstract

Agriculture remains one of the most important sectors in our economy, providing foods, livelihood, and essential raw materials for daily life. However, many farmers continue to face challenges such as limited market access, dependence on middlemen, and lack of real-time weather information, which hinder their productivity and reduce their income. This study aims to develop Agrilink, a mobile platform designed to connect farmers directly to buyers, provide real-time weather updates, and provide direct communication. The system utilizes an application programming interface (API) to enable efficient communication between farmers and buyers, ensuring transparency and fair trade. Through this approach, farmers can sell their products at fair prices, increase their profits, and improve their livelihoods, while consumers benefit from fresher and more affordable goods. The findings emphasize the importance of integrating technology into agriculture to address long-standing challenges, promote sustainability, and enhance economic growth within rural farming communities.

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Chapter 1

Introduction

Farming is one of the important sectors of agriculture that give us food, jobs, and materials for daily life. But many farmers have problems that make their work hard and less profitable. They struggle to find good markets, lack real-time information about the weather, and other income goes to middlemen. Many farmers also do not have modern tools to help them work better. These problems make it difficult for farmers to earn a profitable income, leading to higher market prices and making it harder for people to buy food at fair prices.

According to the GDP of the Philippines, the agriculture sector has always been an important sector of the Philippine economy. It is a significant contributor to the country's GDP. Through the years, however, the percentage share of agriculture in total GDP has declined. From 11.58% contribution to GDP in 2010, this has been reduced to 9.43% in 2015. As regards employment, about 11.29 million Filipinos were engaged in agriculture and fishery production in 2015, representing about 30% of the country's labor force from 2011 to 2015 (DA, 2015). Second, the logistics industry is a fundamental aspect of fulfilling the supply chain in agriculture. Agricultural farmers in the Philippines often use the services of middlemen to transact on their behalf since these intermediaries possess better information, and engaging in logistics affects the marketing margin in the process. Frequently, logistics costs cause marketing margin to increase, leading to high farm-retail prices of agricultural goods. This paper examines if the existence of middlemen improves farmers' wages even with the additional logistics costs incurred. The variables' secondary data were all gathered from the Philippine Statistics Authority, and its annual frequency spanned the period of 1995 to 2019. Using Ordinary Least Squares (OLS) regression analysis procedure and diagnostic tests prove that logistics cost negatively affects marketing margin while farmers' wages have a positive relationship. The findings of this study address the presence of a longer supply chain, asymmetric information, technology, storage and facilities, and added logistics costs in agricultural transactions. Although middlemen have greater market power than the farmers, these mediators are still affected by the changes in the logistics costs since it is unavoidable for them to reduce the price due to the need to competitively sell the commodities.(Quintana, A. A., Chong, M. ., Cordova, M. L., & Camaro,M.A., P. J. . (2022). The Impact of Logistics on Marketing Margin in the Philippine Agricultural Sector. *Journal of Economics, Finance and Accounting Studies* , *3*(2), 300-318).

As stated above, intermediaries have better market information, and their involvement in logistics affects the marketing margin. Although middlemen have greater market power than farmers, they are still impacted by price fluctuations. When product costs increase, middlemen add additional charges to cover their expenses, leading to higher product prices. The study finds that middlemen have a significant impact on the high prices of agricultural products. The goal of this project is to minimize middlemen, provide farmers with direct market access, speed up the buying and selling process, offer real-time weather updates, and facilitate direct communication between farmers and buyers. These objectives aim to empower farmers, improve their livelihoods, and benefit both farmers and consumers.

To address these challenges, the proponents aim to develop a mobile application that will help farmers overcome market difficulties. This application will minimize middlemen and provide direct market access, allowing farmers to sell their products at fair prices while enabling consumers to purchase goods at lower costs. The mobile application will benefit both farmers and consumers. Additionally, it will offer real-time weather updates to help farmers make informed decisions and increase productivity. By addressing these challenges, the application will highlight the importance of technology in improving farmers' lives and ensuring fair product pricing.

Background of the Study

Agriculture has always been the backbone of our economy, supporting communities with food, livelihood, and essential raw materials for daily life. However, farmers frequently face challenges that make it hard for them to earn a fair income. They often rely on middlemen, who take a large portion of the profits and struggle to find markets where they can sell their products directly. These problems not only reduce productivity but also make it harder for farmers to adjust to agricultural needs, which affects their income and quality of life.

Middlemen often control market price, leaving farmers with little bargaining power and trapping them in a cycle of low income. This lack of direct communication between farmers and buyers forces farmers to rely on middlemen to get their products to market, reducing their profits. At the same time, buyers end up paying more because of these middlemen, which makes the process inefficient and increases costs for consumers. Minimizing middlemen can give farmers a chance to sell directly their product to buyers, ensuring that they get a fair share of the profits while buyers enjoy lower costs and fresher, higher-quality products. This direct connection builds trust between farmers and buyers.

Another challenge is that farmers have a lack of access to tools or information that could help them plan their activities effectively. For instance, they often don’t get real-time weather updates, making it hard for them to choose the appropriate time to plant, irrigate, and harvest crops. Without this information, farmers can face unexpected losses. This uncertainty not only affects their earnings but also makes it harder for them to recover from losses, affecting both their financial stability and productivity.

AgriLink seeks to address these challenges and create a farming system that works better for everyone and supports farmers' growth and their success. By providing farmers with the tools to minimize middlemen and sell directly to consumers, while also giving them access to real-time weather updates. Addressing these challenges will empower farmers, improve their livelihoods benefiting both farmers and consumers.

Statement of the problem

This study aims to solve the problems farmers face. These problems make it hard for farmers to earn proper income, sell their products, and make good decisions. By addressing these issues, the study hopes to help farmers improve their work, find better market opportunities, and have a more stable income.

The following challenges that farmers faced are:

- The presence of middlemen reduces farmers' income and increases costs for buyers.
- A lack of market access causes inefficiencies in the supply chain and limits farmers' ability to sell their products at better prices.
- Farmers lack access to real-time weather data, making it difficult to plan their activities effectively.
- Weak access to technology in agriculture, where many farmers are unable to adopt modern tools and platforms that could significantly improve their productivity and decision-making.
- Farmers struggle to recover from losses because they lack sufficient information or money. Also, fluctuating market prices, severe weather, and crop failures make it even more difficult.

Objectives of the Study

The main objective of this study is to develop a system that supports farmers in growing their business. This project will utilize a mobile application using an application programming interface (API) that will allow direct communication between farmers and buyers.

Specifically, this system aims to:

- Provide farmers with better market access to help them sell their products at more affordable prices.

- Minimize middlemen to ensure more profits go to the farmers.
- Provide farmers with real-time weather updates to help them plan better.
- Connect farmers to verified and trusted markets or buyers where they can directly sell their products without the need for middlemen.
- Develop systems that can help farmers grow their business by providing mobile applications that help them to direct market access.

Significance of the Study

This study is significant because it will address the common challenges faced by farmers, such as low income, limited direct access to markets, and reliance on middlemen. The proponents develop a platform that will help farmers to increase their income while minimizing the need for middlemen. By providing farmers with direct access to markets and modern tools, farmers will be able to increase their earnings and improve their quality of life. Buyers will also benefit from having fresher and more affordable products.

In addition to market access, the lack of direct communication between farmers and buyers can also lead to delays in transactions, missed opportunities as well and farmers will not be able to understand what buyers need. This platform will seek to improve the process of buying and selling, making it faster, easier, and improving communication between the farmers and the buyers. Farmers can set their product at the right price so that the market can make the right profit, and also consumers can buy the product at the right price. Minimizing the need for middlemen.

Another challenge this platform will address is the limited access to real-time weather information. By providing farmers with real-time weather updates, they will be able to plan their activities more effectively, reducing risks like crop loss and improving productivity. With access to real-time weather updates, farmers can make informed decisions about when to plant, irrigate, and harvest crops, ensuring quality products and a higher income.

This study aims to improve livelihoods for farmers, enhance the efficiency of the agricultural markets, and promote trust and transparency between farmers and buyers. This study not only aims to ensure that farmers receive fair income for their efforts but also seeks to make quality agricultural products more accessible and affordable for buyers.

By addressing these challenges, this aims to create a positive impact in the agricultural sector, promoting economic growth and demonstrating how technology and innovation can practically solve real-world problems.

Scope and Limitations

The study will focus on addressing the challenges faced by small and medium-scale farmers in rural areas. It primarily targets issues related to market access, middlemen, lack of real-time weather information, and crop recommendation where it provides information per crop. It aims to develop a platform that facilitates direct communication between farmers and buyers, minimizing middlemen, and provides real-time weather updates to enhance agricultural decision-making.

The scope of this study will focus on small and medium-scale farmers in rural areas where access to technology and information is limited. It will not cover other agricultural sectors such as large-scale farming, aquaculture, or forestry. The solutions and findings are designed to address the challenges of rural farming communities; they will not apply to other areas of agriculture. This study will assume that farmers and buyers are willing to use the platform.

Definition of terms

1. **Farmers** - Engaging in the cultivation of crops and managing livestock or other activities that produce food, raw materials, and other goods
2. **Middlemen** - Middlemen between farmers and buyers who facilitate the buying and selling of agricultural products, which usually cause high profits. They may control prices and take the other portion of the profits of the farmers, which reduces their income.

3. **Market Access** - The ability of farmers to reach their buyers or markets where they can sell their products at fair prices. Because limited market access can result in farmers to rely their products on middlemen.

4. **Real-time weather updates** - Are live, continuously updated reports that provide accurate and immediate weather conditions. These updates include data on temperature, rainfall, wind speed, pressure, and other weather variables, delivered as they change. Update to weather data that can help farmers to plan their agricultural activities.

5. **Mobile Platform** - The application designed to facilitate direct communication between farmers and buyers. The goal of this platform is to provide tools for market access, weather updates, and to eliminate middlemen.

6. **Communication** - The process of exchanging ideas, information, thoughts, or feelings between individuals or groups through verbal, non-verbal, or written means.

7. **Agriculture** - The practice of cultivating soil, growing crops, and raising animals for food, fiber, medicinal plants, and other products used to sustain and enhance human life. It involves various activities, including planting, irrigating, harvesting, breeding, and managing resources such as land, water, and equipment.

8. **Application Programming Interface(API)** - is a set of rules and protocols that allows different software applications to communicate with each other.

9. **Front-End** - is the user-facing part of an application or website, involving the design, layout, and interactive elements built using technologies like HTML, CSS, and JavaScript.

Chapter 2

Case Study

Related Literature

Agro Kart - Transforming Agriculture Through Innovative Mobile App Development

"The challenges faced by farmers in the agricultural sector, where they struggle to profit due to reliance on middlemen for selling their goods. The Agro Kart application addresses this issue by enabling farmers to sell directly to consumers, thus improving their earnings and reducing costs for buyers. It features a user-friendly interface and utilizes machine learning to personalize user experiences and product quality assessments. Additionally, the app introduces a sponsorship model, allowing users to financially support farmers, which can enhance their farming operations and profitability"

Sreeja, G., Saraswathi, V., Sathish, V., Swetha, S., & Rajkumar, A. (2023). *Agro Kart – Transforming Agriculture Through Innovative Mobile App Development*. https://doi.org/10.1109/icstcee60504.2023.10584941

This article also relates to the proponent's study because it's all about the challenges faced by farmers where they struggle for their profit because of relying on middlemen for their products. This is related to the proponent project since the proponent uses Mobile applications that would give farmers direct access to the market to sell their products. Improve their earnings and reduce cost for consumers, also want to minimize middlemen. This can contribute to the development of the proposed system, ensuring progress, increased profitability for farmers, and an overall improvement in their livelihoods.

HarvestHub: Mobile Agricultural Markets

"This project aims to develop a web application addressing challenges in agricultural marketing, leveraging technology to connect farmers directly with buyers. Recognizing technology's pervasive role and potential impact, the project seeks to create a user-friendly platform for agricultural trade.The project will employ a comprehensive approach, considering technical development and socio-economic contexts. It begins with analysis of existing systems, informed by literature review and consultations. Iterative cycles of design, implementation, and testing guided by user feedback will be conducted. Key features include support for a local language, secure payments, customizable categorization, and intuitive interface design.The project aims to produce a web application tailored for agricultural marketing. It will provide farmers a platform to showcase products and offer buyers convenient access. By streamlining processes, the application intends to enhance market access, improve transparency, and contribute to sector growth and sustainability. Driven by a commitment to leveraging technology for social good, this project seeks to empower farmers and promote economic development. Through stakeholder collaboration and ongoing evaluation, it aims for sustainability and scalability."

Deepika, G., Begum, H., Reddy, G. N., Joshini, G., Sathyanarayanan, D., & Pallavi, L. (2024). *HarvestHub: Mobile Agricultural Markets*. *4*, 449–454. https://ieeexplore.ieee.org/document/10672970

The article above talked about the overall function of agriculture using web applications. This is related to the proponents' projects that provide farmers a platform to showcase products and offer buyers convenient access. By streamlining processes, the application intends to enhance market access, improve transparency, and contribute to sector growth and sustainability. This Project seeks to empower farmers and promote economic growth and with the help of this mobile application the farmers can easily access the market and provide direct access to farmers and consumers. If this happens, that would be a big help to farmers for their daily lives

Empowering Smallholders and Local Food Markets with Smartphones and Social Networks

"In low income countries, a large majority of the population depends on smallholder farming for their livelihoods and well-being. However, smallholders face significant challenges, often due to a lack of access to information about prices and markets. This paper describes ongoing work on a methodological approach for addressing these challenges through the use of an Android-based mobile-cellular platform application. Using a social networking approach, the application enables smallholders to interact with traders, retailers, consumers, and each other in such a way as to encourage the emergence of sustainable and fair regional markets. It will increase profitability by providing information on price and demand, and enable better cooperation and knowledge-sharing between farmers. This may not only improve yields, but also increase diversity in agricultural products produced: farmers can better adapt to changing climatic conditions and be less vulnerable to volatile commodity prices."

Henriques, J. J., & Kock, B. E. (2012). Empowering Smallholders and Local Food Markets with Smartphones and Social Networks. *Global Humanitarian Technology Conference*, 181–185

The article is related to the proponent's study because both aim to provide an application that connects smallholder farmers with traders, retailers, and consumers to promote fair and sustainable markets. Similar to the proposed system, which offers direct market access, the article highlights the importance of enabling farmers and consumers to communicate without middlemen. This approach helps farmers increase their profits while ensuring fair prices and demand for their products.

Chapter 3

Methodology

This chapter outlines the research methodology that will be used in developing a platform that will help farmers. The main objective of this project is to design an application that will help farmers sell their products in the market without going through middlemen. So that both farmers and consumers can benefit from the fair price. This project will connect farmers to consumers, make buying and selling processes faster, and minimize middlemen, as well as real-time weather updates to help farmers make informed decisions on their activities. This part will help detail the research approach, market problem, direct communication and ensure farmers profit that will be used for this goal.

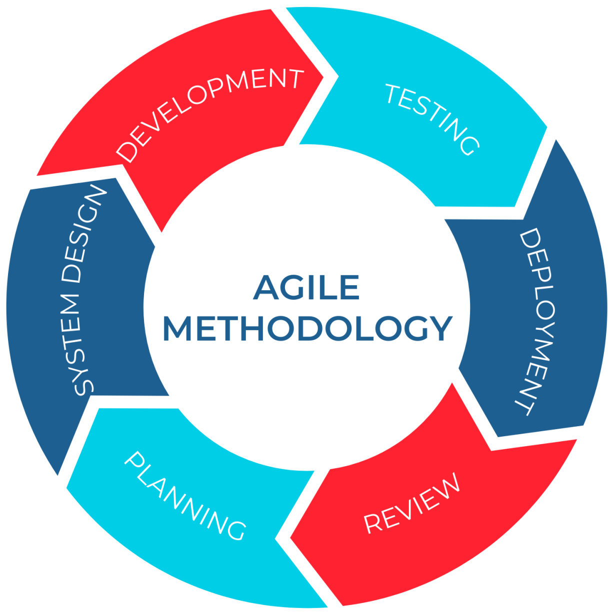
The research will adopt a planned approach to provide the effective development and implementation of the proposed system. At first this platform will be conducted to identify the problems of farmers for better improvements. Also, an effective analysis of the existing agricultural system will be conducted to identify the most needs of farmers and consumers. Based on this review, identify what would be of this platform. Where the architecture and components of the system will be defined.

Research and Development Stage

The research and development stage is a critical phase in the creation of this Agrilink. During this stage, the development process will be based on the challenges faced by farmers and consumers where the farmers are abused by some middlemen who incharge to their products before going to market. These farmers provide valuable inputs on the system design, functionality, and user needs, ensuring that the final product is practical and meets the requirements and needs of farmers and consumers.

Additionally, the R&D stage will begin with an analysis of the data based on the farmers and consumers challenges. These challenges will help to identify their problems so we can find a way to solve them.

Agile Development Methodology



The Software Development life cycle (SDLC) is a structured approach used in the development of the agrilink to develop a project that is completed systematically, efficiently and effectively. The SDLC will guide the development process from the initial planning phase to the final deployment, ensuring that the system meets the needs of the users and aligns with the insights provided by the farmers and consumers.

The SDLC will consist of the following stages:

1. Planning

In the system planning phase, the proponents will look for a problem that needs to be solved in today's time. The researchers brainstormed about what kind of system project to propose and its title and contents. The team gathered data through observation and listed all the suggestions gathered from different audiences and came up with solutions to farmer's daily challenges. Such as, lack of direct communication between farmers and buyers, Real-time weather updates, relying on middlemen that take large portions of their profits. The objective of this system is to bridge the gap between farmers and consumers. Ensuring fair pricing, minimizing middlemen and giving real-time weather information. This phase also involves setting clear goals, defining the project scope.

2. System Design

In the design phase, the system’s architecture and overall structure will be defined based on data collection and observation. The team suggested creating a mobile application. The design will include front-end for user interface ensuring usability for farmers and back-end responsible for core functionality of the mobile app.

Also the system design will use modern technologies such as api for real-time weather updates ensuring farmers can make informed decisions. The design phase will focus on ensuring that the system meets the functionality (enabling direct transactions between farmers and consumers) and non-functionality requirements (scalability, security, and performance) identified in the previous phase.

3. Development

In this stage the proponents developed the proposed project by drafting a system structure, coding and implementation of the system will take place. Based on the design the developer creates software using appropriate programming languages and tools. The clients play an important role by providing feedback on features such as user interface, usability and the performance of the application. And to ensure that the application is user friendly and accessible to the clients.

4. Testing

In this phase, Once the system is developed, the developers test if the system functions and check if all features are working and if the performance of the system is satisfactory for clients. The clients will also be involved in user acceptance testing, where they will test the system in the real-world and identify any issue and if some areas need improvements. Feedback from clients will be helpful in providing what else is missing and needed in the system before it is deployed.

5. Deployment

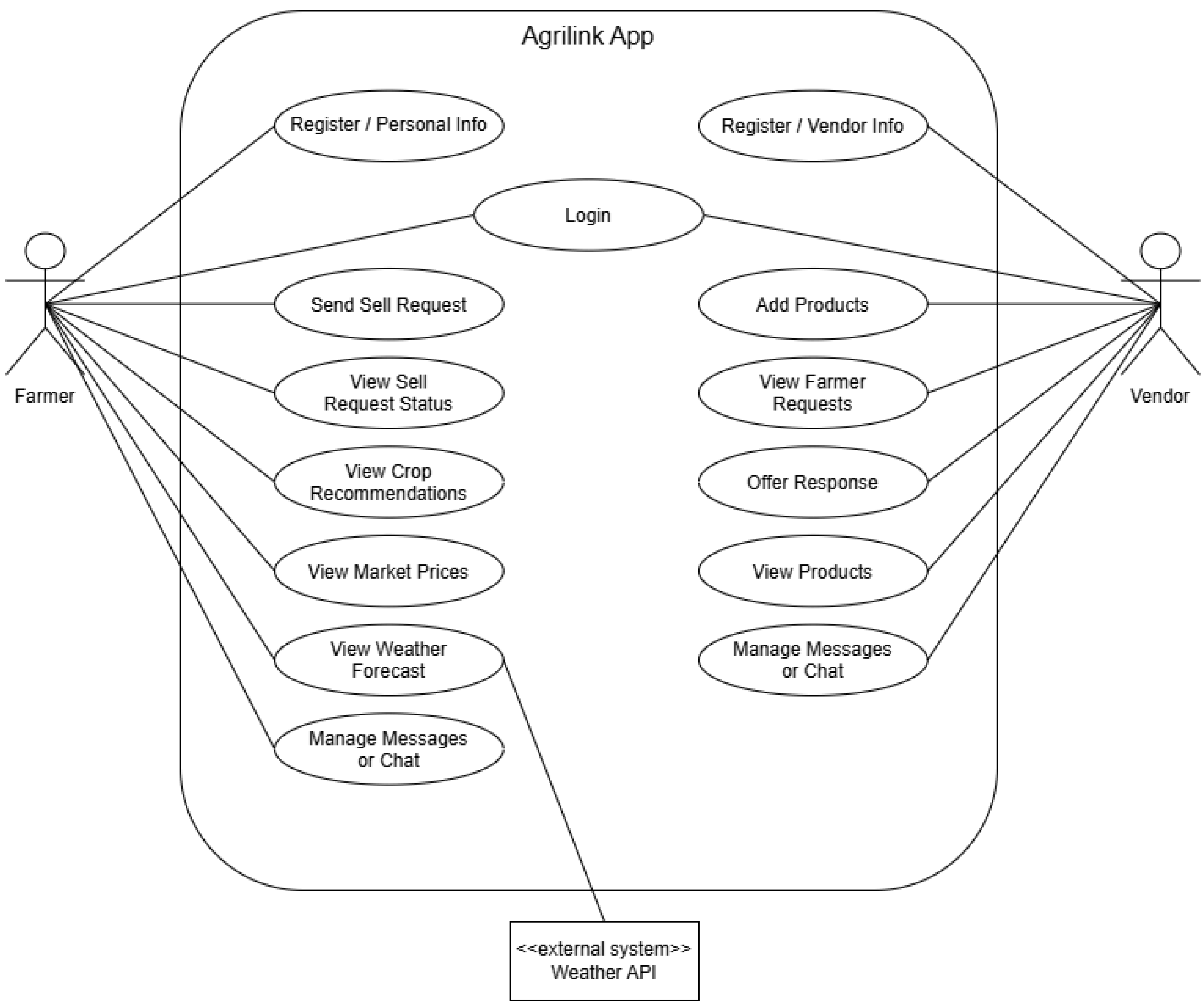
After the successful testing, the system will deploy in markets. This phase will involve installing and configuring the application for actual use, ensuring that it is fully functional and accessible to users. The deployment of this application will also provide tutorials on how to use it, so they can operate effectively. Respondents will be consulted during the deployment process.

6. Review

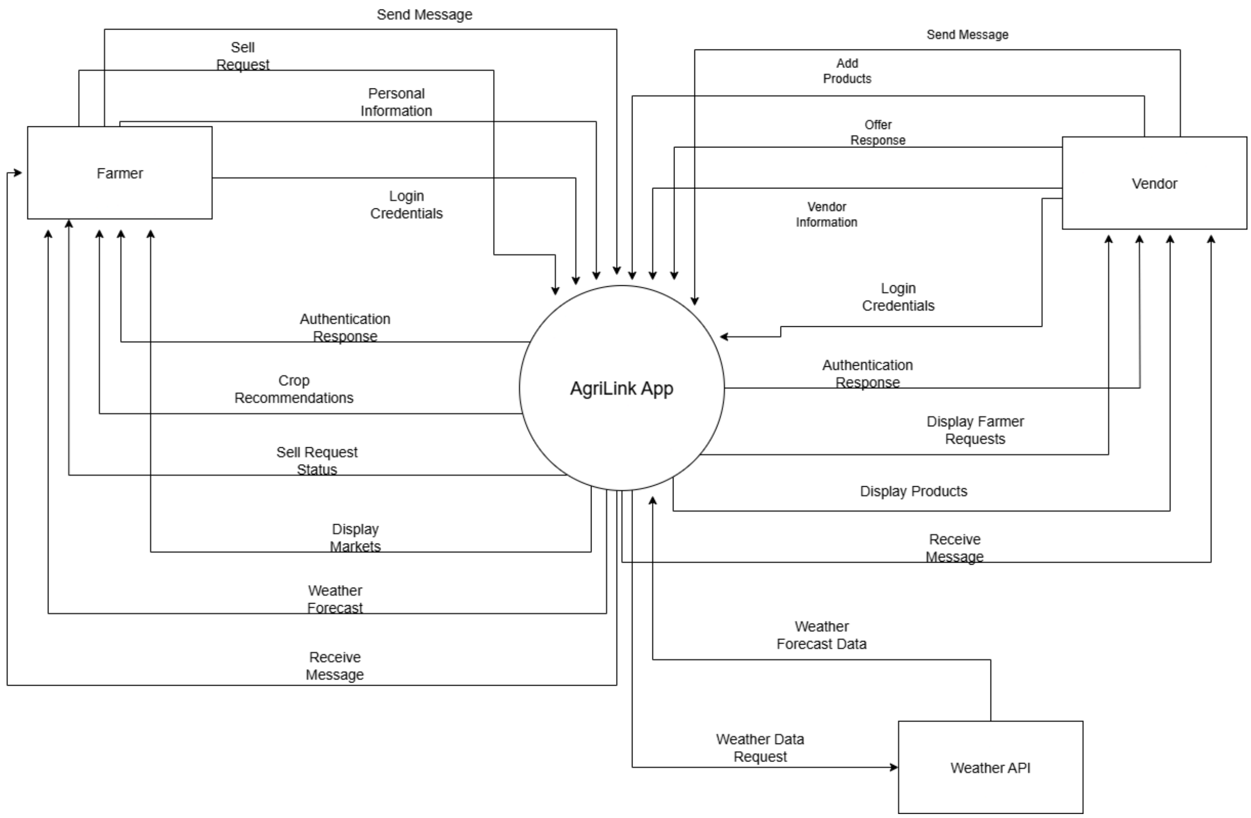
In this stage, clients will provide feedback of the application for its usability, functionality of the system's features and its performance. Helping to identify specific areas needing improvement to align with the clients needs and expectations. The review phase will also assess how well the application meets the objectives outlined in the planning phase, such as enhancing user experience. Issues identified in this phase will be documented for further resolutions, ensuring that the application will deliver its intended purpose to the clients.

Use Case Diagram

The Use Case Diagram is showing how each of the users interacts with the system and what actions they can perform.



Data Flow Diagram



Tree Diagram

```
root
│
├── users
│   ├── farmers
│   │   └── {farmerUID}
│   │       ├── username: "Justine Gomez"
│   │       ├── phone_number: "+639708377465"
│   │       ├── address: "Magsaysay Road, La Paz, Iloilo City"
│   │       ├── password: "Jus123"
│   │       └── userRole: "farmer"
│   │
│   └── vendors
│       └── {vendorUID}
│           ├── username: "Maykel Vendor"
│           ├── phone_number: "+639101261494"
│           ├── address: "La Paz, Iloilo City"
│           ├── password: "vendor123"
│           └── userRole: "vendor"
│
├── markets
│   └── {vendorUID}
│       ├── marketName: "Maykel"
│       ├── barangay: "La Paz"
│       ├── municipality: "Iloilo City"
│       ├── latitude: "10.7096264"
│       ├── longitude: "122.5676698"
│       ├── phoneNumber: "09101261494"
│       ├── street: "La Paz"
│       └── infoComplete: true
│
├── vendor_products
│   └── {vendorUID}
│       └── {productID}
│
```

```
│           ├── vendor_product_name: "Banana"
│           ├── vendor_product_price: 50
│           └── product_unit: "kg"
│
└── requests
    └── {vendorUID}
        └── {requestID}
            ├── farmerUID: "+639708377465"
            ├── productName: "Banana"
            ├── price: 25
            ├── quantity: 50
            ├── paymentMethod: "Cash on Delivery"
            ├── pickupOption: "Farmer will deliver the product"
            ├── deliveryDate: "September 23, 2025"
            ├── requestDate: "September 21, 2025"
            └── status: "Pending"
```

Data collection techniques

In order to gather the necessary information for developing the system, a variety of data collection techniques will be employed. These techniques will help the system meet the needs of users, particularly in both farmers and buyers, while also addressing the current challenges of the farmers. The following data collection methods will be used:

1. **Surveys**

The team uses surveys for gathering data from respondents, including farmers and buyers. These surveys will be used to get information about any problems that farmers face when it comes to sell their products in the market as well as to consumers. Also the question in this survey will be to gather data and specific feedback to solve the farmer's problem when it comes to selling their products and mobile applications development requirements. These will provide valuable insight into the features that should be prioritized to develop mobile applications to help farmers and buyers.

2. **Interviews**

Interviews will be conducted with selected respondents to collect data and information. These respondents will be chosen based on their experience in the market. The interviews will allow for deeper understanding of the challenges faced by farmers and consumers, their workflow, and how a mobile application could improve their efficiency and productivity. During the interviews, open ended questions will encourage detailed responses, offering insights that may not be captured through survey alone.

3. **Observations**

The Proponents conducted observations to understand the daily problems faced by farmers. Farmers often have a hard time selling their products because they rely on middlemen. And many farmers don't have direct access to markets, in addition to that sudden change in weather conditions can also affect farmers planting and harvesting crops. By observing farmers the proponents will gain a better understanding on what platform would best meet farmers' needs.

Survey Questionnaire for System Assessment

This survey aims to gather information from farmers to better understand the challenges they faced and guides us in creating a platform that connects farmers directly to local markets, provides real-time weather updates, and helps farmers to have a better income.

Demographic Information

1. **How do you currently sell your agricultural products?**

   - ☐ Through middlemen
   - ☐ Direct to buyers
   - ☐ Local markets
   - ☐ Online
   - ☐ etc

2. **How easy is it for you to find buyers for your products?**

   - ☐ Very Easy

- ☐ Easy
- ☐ Neutral
- ☐ Difficult
- ☐ Very Difficult

3. **How often do you face challenges in transporting your products to the market?**
   - ☐ Frequently
   - ☐ Occasionally
   - ☐ Rarely
   - ☐ Never
4. **Do you have access to a list of markets where you can sell your products?**
   - ☐ Yes
   - ☐ No
5. **How satisfied are you with the profits you make from selling your products?**
   - ☐ Very Satisfied
   - ☐ Satisfied

- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

6. **What factors influence your decision on where to sell your agricultural products?**
   - ☐ Price offers
   - ☐ Convenience
   - ☐ Transportation Availability
   - ☐ Buyer trust
7. **How transparent do you think middlemen are in pricing your products?**
   - ☐ Very transparent
   - ☐ Somewhat transparent
   - ☐ Neutral
   - ☐ Not transparent
   - ☐ Not transparent at All
8. **How often do you rely on middlemen to sell your products?**

☐ Always

☐ Often

☐ Sometimes

☐ Rarely

☐ Never

9. **Do you feel that middlemen take advantage of your earnings?**

☐ Yes

☐ No

10. **Have you ever attempted to sell your products directly to buyers?**

☐ Yes

☐ No

11. **Do you currently use any tools or resources to check the weather before planting or harvesting?**

☐ Yes

☐ No

12. **How do sudden weather changes impact your crops?**

☐ Reduced yield

☐ Crop damage

☐ Delayed harvesting

☐ No impact

13. How often has unpredictable weather negatively impacted your farming activities?

☐ Frequently

☐ Occasionally

☐ Rarely

☐ Never

14. Would access to real-time weather updates help you make better farming decisions?

☐ Yes

☐ No

15. Which weather-related challenges do you face most frequently?

☐ Rainfall

☐ Drought

☐ Storm

☐ etc

16. **Have you ever lost a significant portion of your crops due to extreme weather conditions?**

☐ Yes

☐ No

17. **How do you usually find out about upcoming weather conditions?**

☐ Radio

☐ TV

☐ Mobile apps

☐ etc

18. **Would you be interested in using a platform that connects you directly to local markets?**

☐ Yes

☐ No

19. **How important is a user-friendly design when using technology for farming?**

☐ Very Important

☐ Important

☐ Neutral

☐ Not Important

20. If a platform provided market contacts and real-time weather updates, how likely would you be to use it?

☐ Very Likely

☐ Likely

☐ Neutral

☐ Unlikely

☐ Very Unlikely

Chapter 4

Result and Discussion

Systems and Features

Agrilink is mobile applications that helps farmers sell their product in the market this systems features include:

1. Farmer to market features
   - Connects farmers directly with buyers, enabling them to sell their products at fair prices and reduce reliance on middlemen
2. Weather Forecast Features
   - Provide real-time weather updates to help farmers plan and manage their agricultural activities effectively.
3. Crop recommendations Features
   - Provide farmers with data suggestions on the best crops to plant based on the months they grow fastest, weather conditions, and locations.
4. Call Features
   - Provide a direct Communication allowing users to manage, control, and improve the quality of their transaction.

5. Message Features

    - Allowing users to send, receive, and manage information quickly and effectively.

Implementation Challenges

During the implementation of Agrilink "Empowering farmers through smart technology," the team encountered many challenges to build this application, from suggesting and planning to working on this project. One major challenge was deciding what features farmers needed most, such as helping them sell their products directly to the market and connect with vendors without middlemen. The team also found it difficult to make the app easy to use for farmers who are not familiar with technology, and poor internet connection in some areas also made it harder to test and use the app smoothly.

There are some challenges that farmers and vendors face in Barangay Trapiche, Oton, during the implementation. Some farmers were not familiar with using applications or smartphones. Some had difficulty installing the app, navigating its features, or understanding the functions. Another challenge was the poor internet connection in some areas, which made it difficult to install the app. These challenges are the same for the vendors in the Oton public market; some of them have no internet connection, some have difficulty using the application consistently, and some have limited knowledge about smartphones.

Despite these challenges, the implementation of AgriLink demonstrated that farmers are open to adopting digital solutions when proper training, guidance, and support are provided. With continuous education, improved internet infrastructure, and further system enhancements, these issues can be minimized, allowing AgriLink to become an even more effective tool for empowering farmers and improving their productivity.

User Feedback

Based on the observation on deploying this application many users have been satisfied with the features of this application.

Key findings:

1. Farmers are very happy with the idea of directing them to market for faster selling of their products
2. Farmers are impressed with the design of the mobile app and believe it will help them sell their products faster without relying on middlemen, allowing them to earn a fair income.
3. Farmers say the app is very helpful because it gives them confidence to sell their goods at the right price, preventing middlemen from taking advantage of them.
4. Vendors appreciated the app because it can help them to buy products directly from farmers fresh at an affordable price.

5. Users found the interface simple, organized, and easy to navigate.
6. Others recommended improving the loading speed, especially in areas with poor internet connection.

Comparison to other System

Compared to other mobile applications, Agrilink aims to meet the needs of local farmers by providing practical easy to use features. While the Agro Kart application focuses on helping farmers sell their products directly to consumers using machine learning and sponsorship models to increase profits (Sreeja et al., 2023), AgriLink offers a more practical and community based approach. Instead of being direct to consumers, AgriLink provides a way for farmers to directly connect with market vendors to sell their products.

AgriLink also integrates features such as real time weather updates, message feature, call function for easy communication between farmers and vendors. These additional features make AgriLink accessible and user friendly, especially for local farmers who often don’t have access to technology.

Chapter 5

Summary, Conclusion, and Recommendation

Summary

This study aims to design and develop AgriLink "Empowering Farmers Through Smart Technology," a system that directly connects farmers to markets, provides real-time weather information, and offers crop recommendations based on the season. The platform includes features such as farmer-to-market access, weather forecasts, crop recommendations, and call and messaging features to enable direct communication between farmers and vendors. The project followed a structured development process, including requirement analysis, system design, development, testing, and evaluation.

The results showed significant improvements in AgriLink efficiency and user satisfaction. The system successfully met its objectives and proved to be an effective solution for helping farmers sell their products more easily and efficiently.

Conclusion

The project AgriLink: Empowering Farmers Through Smart Technology was developed to address farmers' limited access to markets, reliance on middlemen, and lack of real time weather information. These issues make it hard for farmers to earn fair income, sell their products at low prices, and plan their activities.

The AgriLink system was able to solve that problem by providing a mobile platform that connects farmers directly to markets instead of relying on middlemen. Farmers can easily view available markets, access vendor information and communicate with vendors directly. It also provides real time weather update and crops recommendation based on the season, helping farmers make better planting and selling decisions

By connecting farmers directly to the markets, the system helps reduce the farmers' reliance on middlemen on selling their products at lower prices.

Recommendations

Based on the development and evaluation of the project, the following recommendations are made:

1. **Smart Crop Recommendation -** Include description of each crop once clicked. For better information that can enhance farmers' knowledge about crops.
2. **Enhance App Performance** - Optimize the loading speed and overall system performance to ensure smooth usage, especially in areas with poor connectivity.
3. **Offline Accessibility -** Add features that allow users to access essential information, such as product listings and product updates, even without an internet connection

4. **User Training and Support -** Provide a simple guide or tutorials to help farmers and vendors fully understand and use all the features of the app.
5. **Improve Interface Design** – Continue Enhancing the app's interface to keep it simple, organized, and user-friendly for all age groups and experience levels.
6. **Expand Market Network -** Integrate more verified vendors to give farmers wider opportunities to sell their products directly.
7. **Feedback System -** Include an in-app feedback feature so users can easily share their experience and suggestions for continuous improvement.
8. **Notification Alerts** – Add real-time notification for market updates, buyer inquiries, and weather alerts to keep users informed and engaged.

Final Thoughts

The development of Agrilink demonstrates the significant potential of technology in transforming the agricultural sector and improving the lives of farmers. By providing direct market access, real-time weather updates, and easy communication between farmers and vendors, the system reduces the dependence on middlemen and promotes fair trade. Through this project, the team realized that even simple technological solutions can make a difference in addressing real-world problems. Agrilink is not just an application, it is a step toward empowering farmers, improving productivity, and promoting a more sustainable and connected farming community.

Reference (APA format)

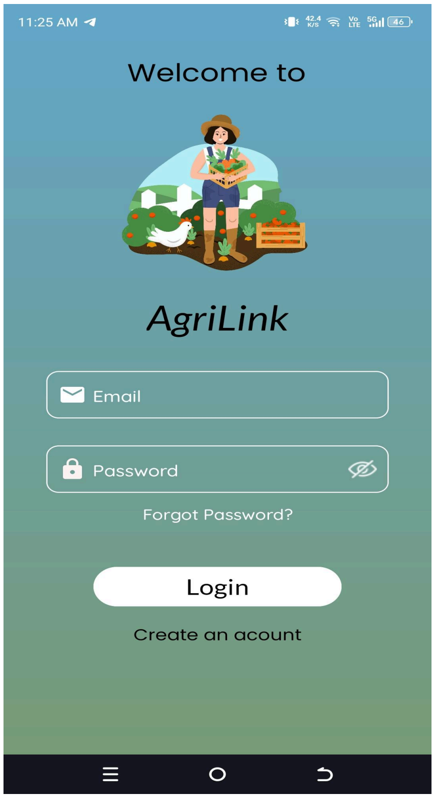
1. Sreeja, G., Saraswathi, V., Sathish, V., Swetha, S., & Rajkumar, A. (2023). *Agro Kart – Transforming Agriculture Through Innovative Mobile App Development*. https://doi.org/10.1109/icstcee60504.2023.10584941

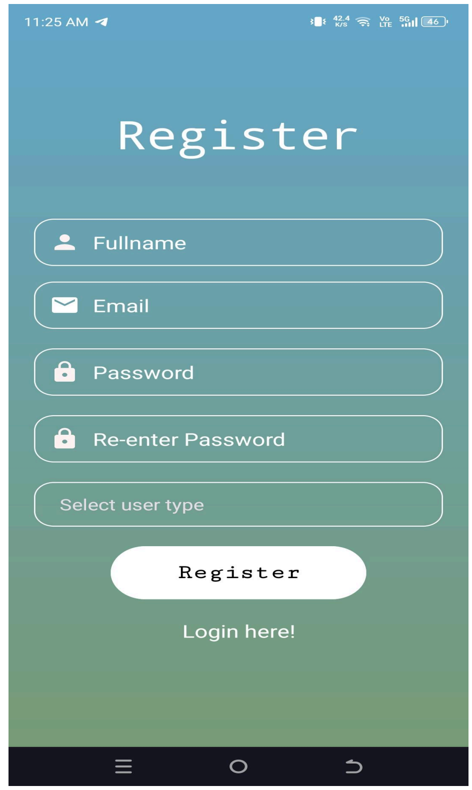
2. Deepika, G., Begum, H., Reddy, G. N., Joshini, G., Sathyanarayanan, D., & Pallavi, L. (2024). *HarvestHub: Mobile Agricultural Markets*. *4*, 449–454. https://ieeexplore.ieee.org/document/10672970

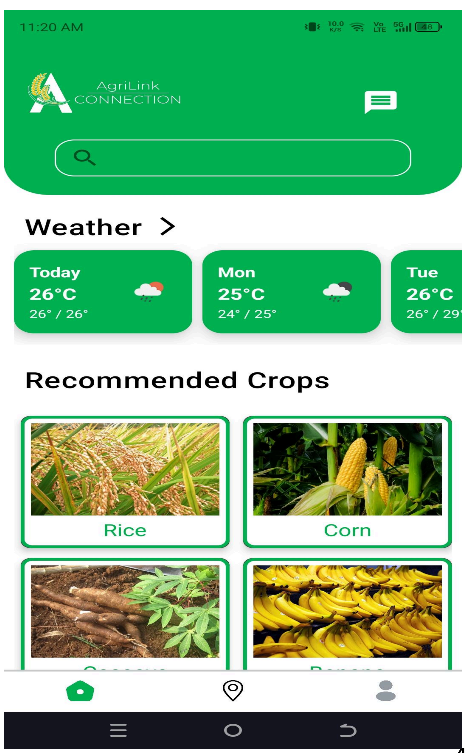
3. Henriques, J. J., & Kock, B. E. (2012). Empowering Smallholders and Local Food Markets with Smartphones and Social Networks. *Global Humanitarian Technology Conference*, 181–1

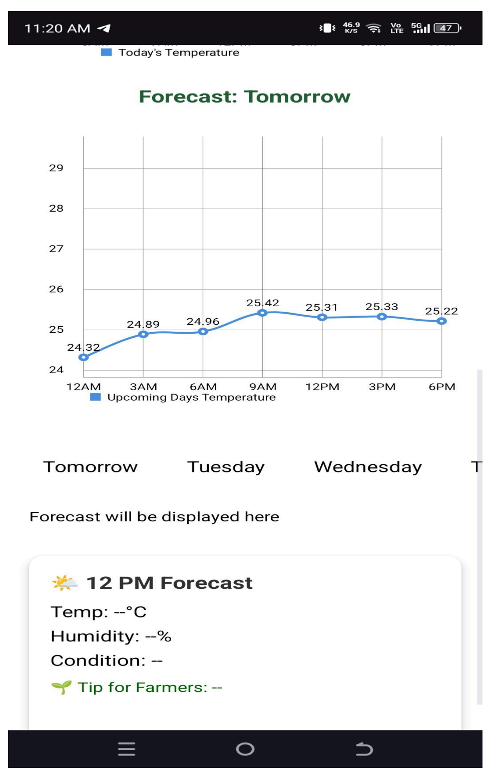
Appendices

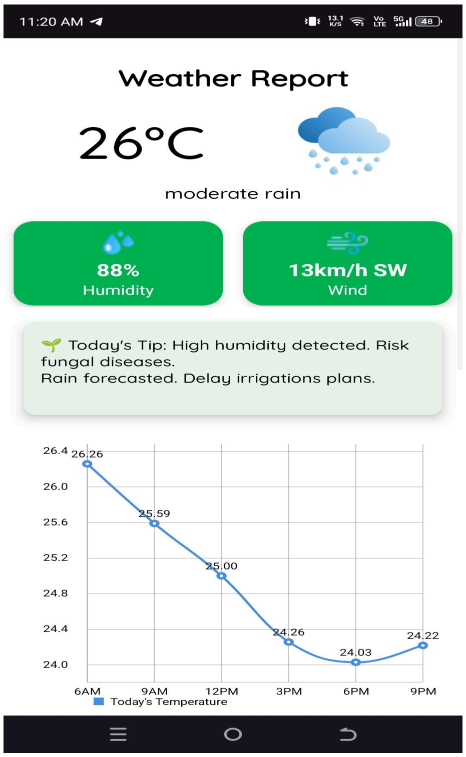
Screenshots - Farmers Side

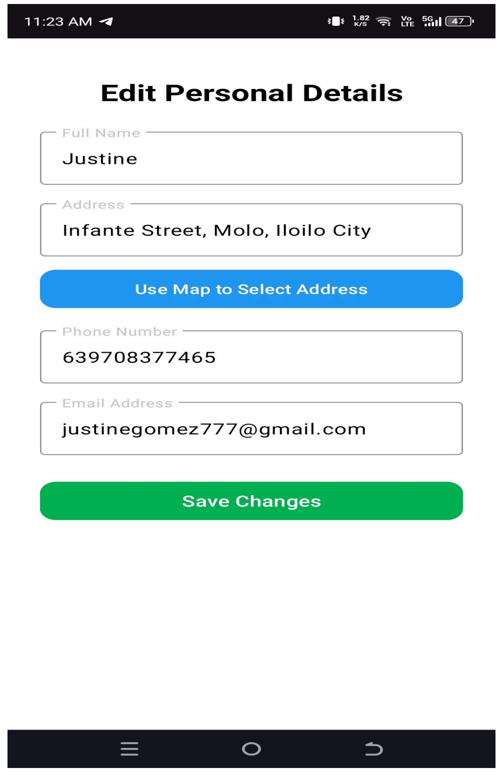


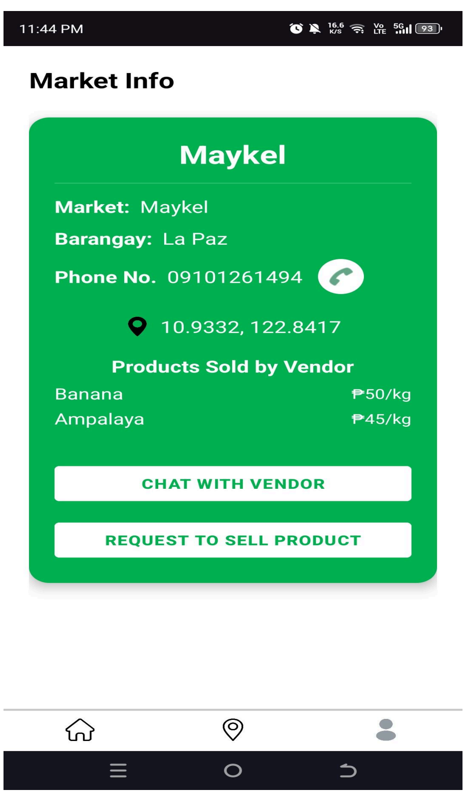


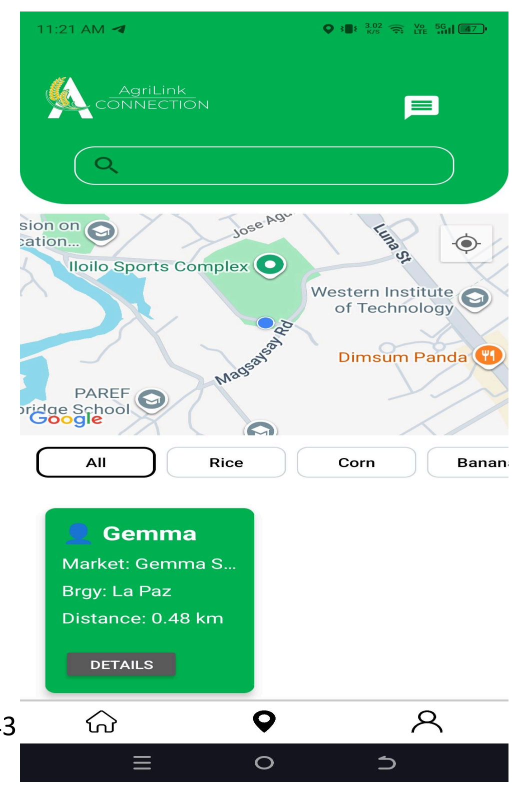


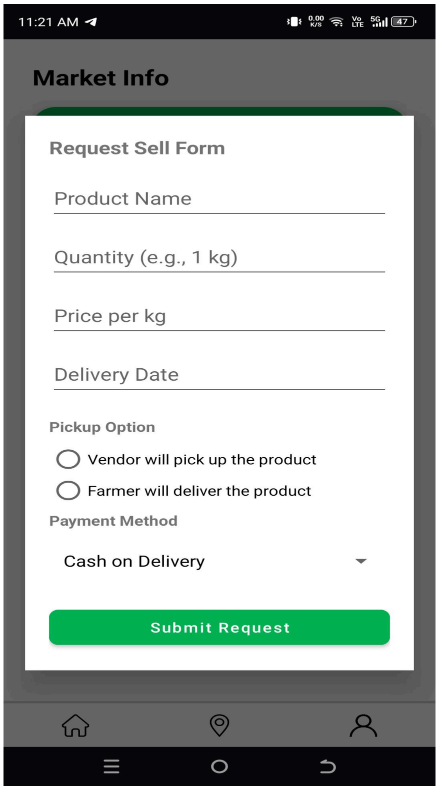


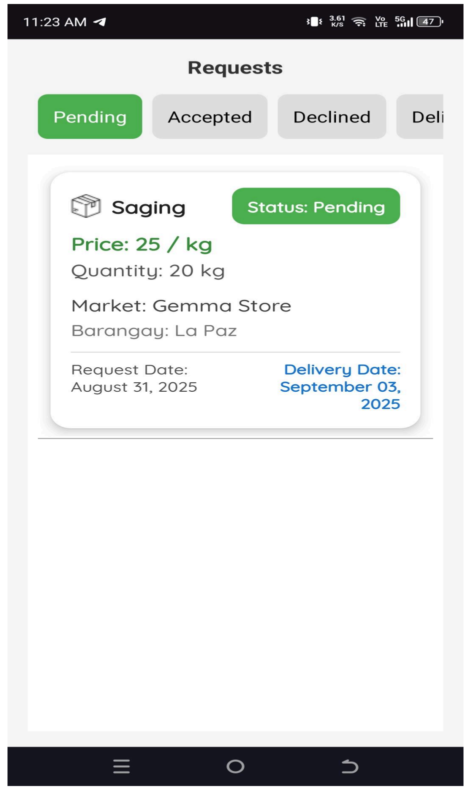

11:20 AM
Weather Report
26°C
moderate rain
88%
Humidity
13km/h SW
Wind
Today's Tip: High humidity detected. Risk fungal diseases.
Rain forecasted. Delay irrigations plans.
26.4
26.0
25.6
25.2
24.8
24.4
24.0
26.26
25.59
25.00
24.26
24.03
24.22
6AM
9AM
12PM
3PM
6PM
9PM
Today's Temperature


11:23 AM
Edit Personal Details
Full Name
Justine
Address
Infante Street, Molo, Iloilo City
Use Map to Select Address
Phone Number
639708377465
Email Address
justinegomez777@gmail.com
Save Changes


11:44 PM
Market Info
Maykel
Market: Maykel
Barangay: La Paz
Phone No. 09101261494
10.9332, 122.8417
Products Sold by Vendor
Banana ₱50/kg
Ampalaya ₱45/kg
CHAT WITH VENDOR
REQUEST TO SELL PRODUCT

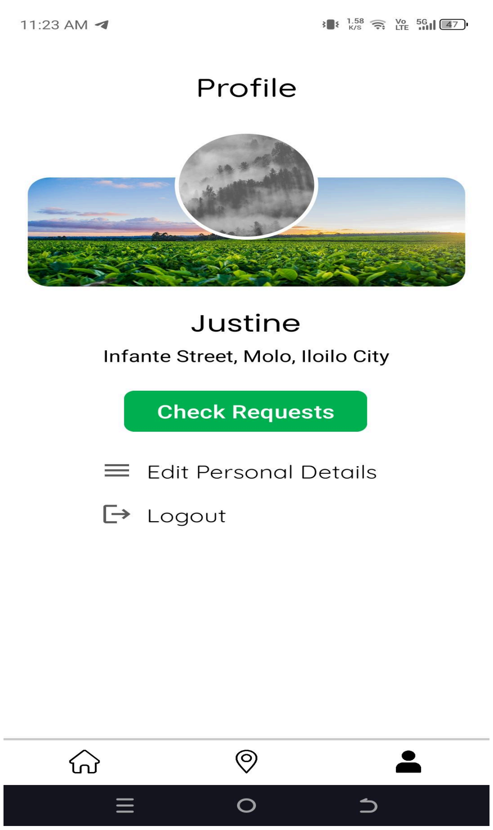

11:21 AM
AgriLink
CONNECTION
Iloilo Sports Complex
Jose Agu
Luna St
Western Institute
of Technology
Dimsum Panda
Magsaysay Rd
PAREF
Google
All
Rice
Corn
Banan
Gemma
Market: Gemma S...
Brgy: La Paz
Distance: 0.48 km
DETAILS

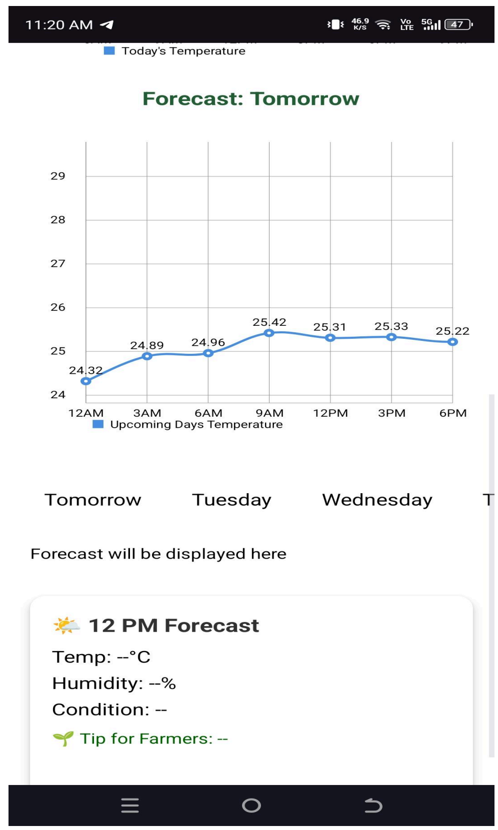

11:21 AM
Market Info
Request Sell Form
Product Name
Quantity (e.g., 1 kg)
Price per kg
Delivery Date
Pickup Option
Vendor will pick up the product
Farmer will deliver the product
Payment Method
Cash on Delivery
Submit Request

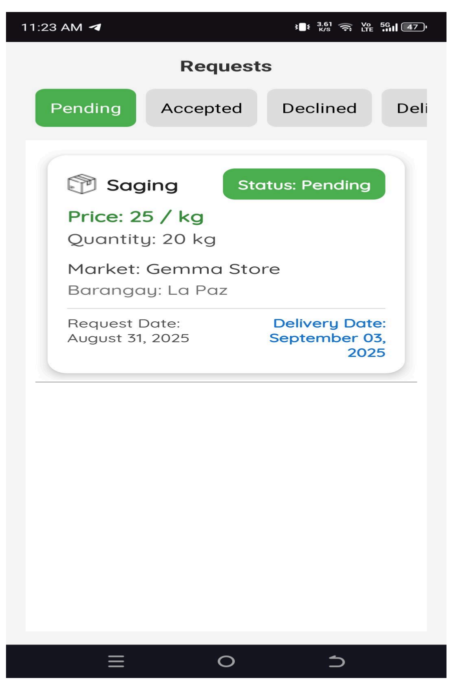

11:23 AM
Requests
Pending
Accepted
Declined
Deli
Saging
Status: Pending
Price: 25 / kg
Quantity: 20 kg
Market: Gemma Store
Barangay: La Paz
Request Date:
August 31, 2025
Delivery Date:
September 03, 2025


11:21 AM
Gemma
Goodmorning
Type a message...

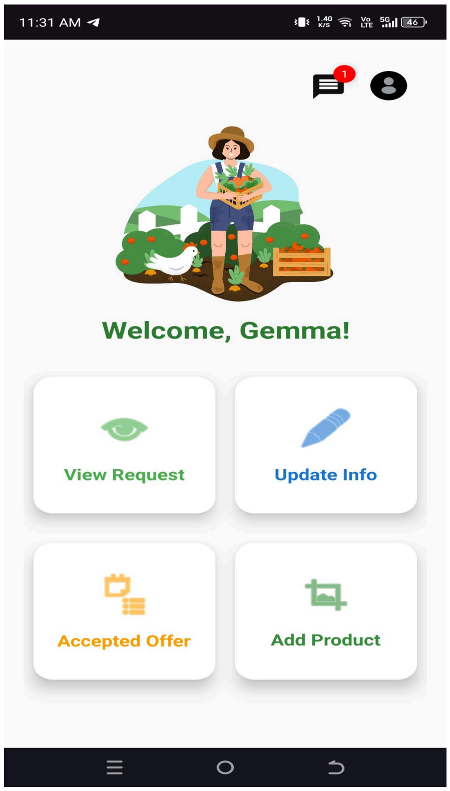

11:23 AM
Gemma
Goodmorning
11:21 AM

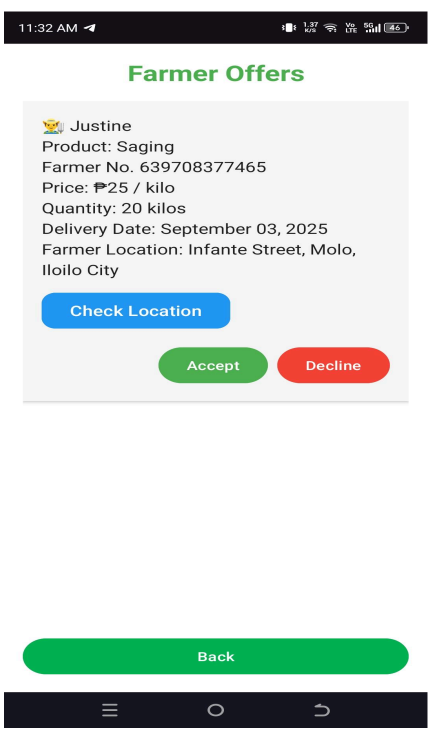

11:23 AM
Profile
Justine
Infante Street, Molo, Iloilo City
Check Requests
Edit Personal Details
Logout

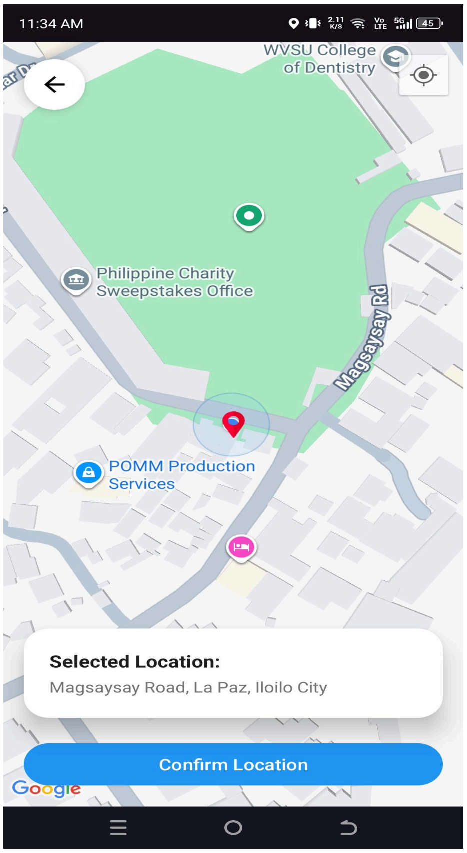

11:20 AM
Today's Temperature
Forecast: Tomorrow
29
28
27
26
25
24
24.32
24.89
24.96
25.42
25.31
25.33
25.22
12AM
3AM
6AM
9AM
12PM
3PM
6PM
Upcoming Days Temperature
Tomorrow
Tuesday
Wednesday
Forecast will be displayed here
12 PM Forecast
Temp: --°C
Humidity: --%
Condition: --
Tip for Farmers: --

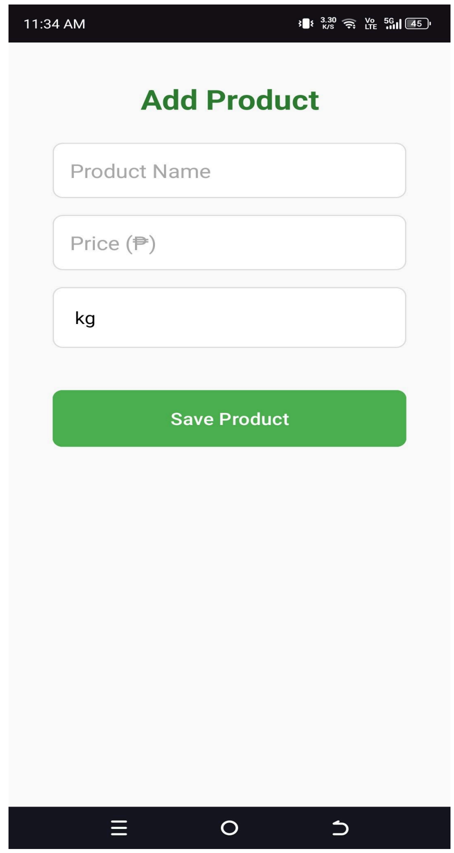

11:23 AM
Requests
Pending
Accepted
Declined
Saging
Status: Pending
Price: 25 / kg
Quantity: 20 kg
Market: Gemma Store
Barangay: La Paz
Request Date:
August 31, 2025
Delivery Date:
September 03, 2025

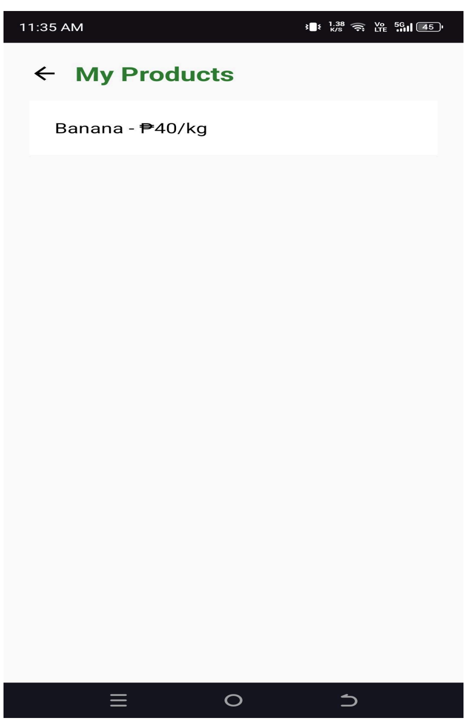
Screenshots - Vendors Side

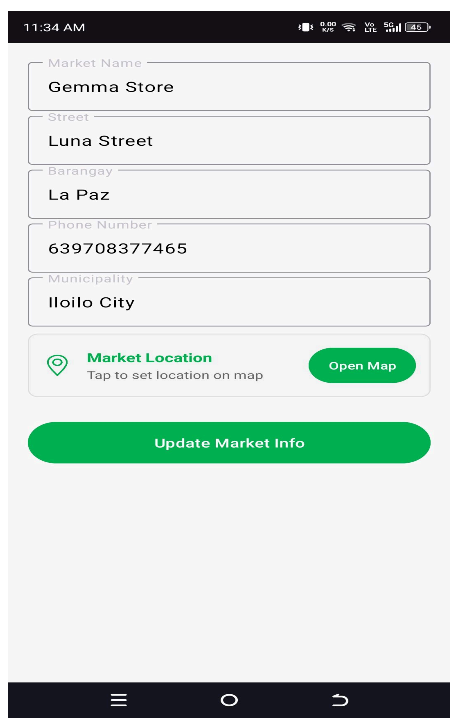




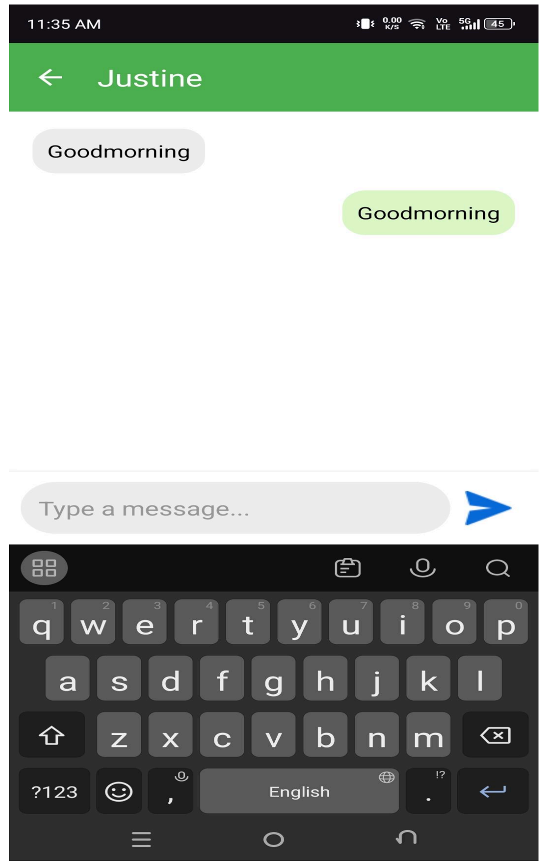

11:34 AM
WVSU College
of Dentistry
Philippine Charity
Sweepstakes Office
Magsaysay Rd
POMM Production
Services
Selected Location:
Magsaysay Road, La Paz, Iloilo City
Confirm Location
Google


11:34 AM
Add Product
Product Name
Price (₱)
kg
Save Product


11:35 AM
My Products
Banana - ₱40/kg


11:34 AM
Market Name
Gemma Store
Street
Luna Street
Barangay
La Paz
Phone Number
639708377465
Municipality
Iloilo City
Market Location
Tap to set location on map
Open Map
Update Market Info


11:35 AM
Justine
Goodmorning
11:21 AM
1


11:35 AM
Justine
Goodmorning
Goodmorning
Type a message...

PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

Curriculum vitae

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Nationality: Filipino

Languages: English, Filipino

Educational Background

Bachelor of Science and Information Technology

PHINMA University of the Iloilo - Iloilo City

2022 – 2026

Senior High School

Lawigan National High School, Iloilo City

2016 – 2021

Skills

- editor and encoder
- Time management and teamwork

Curriculum vitae

Name: Justine Gomez
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Place of Birth: San Fernando, Bauang, La Union
Civil Status: Single
Nationality: Filipino
Languages: English, Filipino

Educational Background

Bachelor of Science and Information Technology
PHINMA University of the Iloilo - Iloilo City
2022 – 2026

Senior High School
Roberto H. Tirol NHS - Concepcion Iloilo
2016 – 2021

Skills

- Programming
- Problem Solving

Curriculum vitae

Name: Christian Dave Yanila
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Personal Information

Date of Birth: January 22, 2004
Place of Birth: Languanan, San Joaquin, Iloilo
Civil Status: Single
Nationality: Filipino
Languages: English, Filipino

Educational Background

Bachelor of Science in Information Technology

PHINMA University of Iloilo | 2022 - 2026

Junior Highschool - Senior Highschool

Strand OF TVL (ICT) Information and Communication Technology

Oton National High School | 2016 - April 2022

Skills

- Programming
- Problem Solving